

# Year 11    Mathematics Extension II    Term 4    Assessment    2009

## Question 1:    (15 Marks)    (START A NEW PAGE)

**Marks**

- (a) Given that  $z = 2 + i$  and  $w = 1 + 3i$ , express the following in the form  $z = x + iy$  where  $x$  and  $y$  are real numbers.
- (i)  $z + \bar{w}$  2
- (ii)  $z^2$  2
- (iii)  $\frac{z}{w}$  2
- (b) Given that  $z = 2 + 3i$  is a root of the equation  $z^3 - 2z^2 + pz + q = 0$ , where  $p$  and  $q$  are real.
- (i) Find the other two roots of the equation. 2
- (ii) Find the values of  $p$  and  $q$ . 2
- (c) (i) If  $z = \cos \theta + i \sin \theta$ , use De Moivre's Theorem to prove that  $z^n + z^{-n} = 2 \cos n\theta$ . 2
- (ii) Hence deduce that  $\cos \theta \cos 2\theta = \frac{1}{2}(\cos 3\theta + \cos \theta)$ . 3

## Question 2:    (15 Marks)    (START A NEW PAGE)

**Marks**

- (a) (i) Solve  $\alpha^2 = 5 - 12i$ , expressing both answers in the form  $a + ib$  where  $a$  and  $b$  are real numbers. 3
- (ii) Hence solve  $z^2 - (1 - 4i)z - (5 - i) = 0$ , expressing both answers in the form  $x + iy$  where  $x$  and  $y$  are real numbers. 3
- (b) Sketch, on an Argand Diagram, the intersection of the regions  $|z| \geq 4$  and  $0 \leq \arg(z) \leq \frac{\pi}{4}$ . 3
- (c) If  $z = \frac{1+i}{1-i}$  and  $w = \frac{\sqrt{2}}{1-i}$  are two complex numbers.
- (i) Find the modulus and arguments of  $z$  and  $w$ . 2
- (ii) Plot the points representing  $z$ ,  $w$  and  $z + w$  on an Argand Diagram. 2
- (iii) Hence show that  $\tan\left(\frac{3\pi}{8}\right) = 1 + \sqrt{2}$  2

**Question 3: (15 Marks) (START A NEW PAGE)****Marks**

- (a) (i) Express  $1 + i\sqrt{3}$  in mod-arg form. 2
- (ii) Hence evaluate  $(1 + i\sqrt{3})^8$ , expressing your answer in the form  $x + iy$  where  $x$  and  $y$  are real numbers and not trigonometric expressions. 2
- (b) Solve  $\bar{z} - \frac{12}{z} = 1 - i$ , expressing your answer in the form  $a + ib$  where  $a$  and  $b$  are real numbers. 4
- (c) The point  $Z$  represents the complex number  $z$  in the complex plane. If  $\operatorname{Re}\left(\frac{z - ib}{z - a}\right) = 0$ , where  $a$  and  $b$  are real, prove that the locus of  $Z$  is a circle and give its centre and radius. 4
- (d) If  $z = r(\cos \theta + i \sin \theta)$ , prove that  $\frac{z}{z^2 + r^2}$  is real and express its value in terms of  $\theta$ . 3

**Question 4: (15 Marks) (START A NEW PAGE)****Marks**

- (a) If  $\alpha$  and  $\beta$  are two complex numbers ( $\alpha \neq \beta$ ) and  $|\alpha| = 1$ , show that  $\left| \frac{\alpha\bar{\beta} - 1}{\alpha - \beta} \right| = 1$ . 3
- (b) (i) The points  $P$ ,  $Q$  and  $R$  represent the complex numbers  $z$ ,  $w$  and  $z + w$  respectively on an Argand diagram, where  $\arg(w) > \arg(z)$  and  $0 < \arg(z) < \frac{\pi}{2}$ . Plot the positions of points  $P$ ,  $Q$  and  $R$  on an Argand diagram. 1
- (ii) If  $O$  is the origin, what type of quadrilateral is  $OPRQ$  given that  $\left| \frac{z - w}{z + w} \right| = 1$ ? 1
- (iii) Hence, what can be stated about the expression  $\frac{w}{z}$ ? 2
- (c) (i) Write  $-1$  in mod-arg form and hence express both complex roots of  $z^3 = -1$  in mod-arg form. 3
- (ii) If  $z = \lambda$  is either of the two complex roots of  $z^3 = -1$ , prove that  $\lambda^2 - \lambda + 1 = 0$ . 2
- (iii) Evaluate  $(\lambda^2 + \lambda + 1)^8 + (\lambda^2 + \lambda - 1)^8$ . 3

**THIS IS THE END OF THE EXAMINATION PAPER**

Alternative

**Question 4: (15 Marks) (START A NEW PAGE)**

**Marks**

- (a) Find the greatest value of  $|z|$  if the complex number  $z$  satisfies the equation  $\left|z - \frac{4}{z}\right| = 8$ . **4**
- (b) (i) Find the two complex cube roots of unity expressing them in mod-arg form. **3**
- (ii) If  $\omega$  is one of the complex cube roots of unity prove that  $1 + \omega + \omega^2 = 0$ . **2**
- (iii) Evaluate  $(1 + \omega - \omega^2)^8 + (1 - \omega + \omega^2)^8$ . **3**
- (iv) Prove that if  $n$  is an integer, then  $1 + \omega^n + \omega^{2n} = 3$  or  $0$ , depending on whether  $n$  is or is not a multiple of 3. **3**

**Extra Questions**

- (a) On the Argand Diagram provided and with the aid of ruler, protractor and a pair of compasses, show the position of the complex numbers represented by
- (i)  $\bar{z}_1$  **1**
- (ii)  $-z_2$  **1**
- (iii)  $z_1 - z_2$  **1**
- (iv)  $iz_1$  **1**
- (v)  $iz_1 + z_2$  **1**

Clearly show all relevant lengths and angles.

If  $\omega$  is a complex root of  $\omega^8 = 1$ , prove that  $\omega + \omega^7$  is real.

# Year 11    Mathematics Extension II    Term 4    Assessment    2009

## SOLUTIONS

### Question 1: (15 Marks) (START A NEW PAGE)

**Marks**

(a) Given that  $z = 2 + i$  and  $w = 1 + 3i$ , express the following in the form  $z = x + iy$  where  $x$  and  $y$  are real numbers.

(i)  $z + \bar{w} = (2 + i) + (1 - 3i)$  2  
 $= 3 - 2i$

(ii)  $z^2 = (2 + i)^2$  2  
 $= 4 + 4i - 1$   
 $= 3 + 4i$

(iii)  $\frac{z}{w} = \frac{2 + i}{1 + 3i} \times \frac{1 - 3i}{1 - 3i}$  2  
 $= \frac{(2 + i)(1 - 3i)}{1 + 9}$   
 $= \frac{2 - 6i + i + 3}{10}$   
 $= \frac{1}{2} - \frac{1}{2}i$

(b) Given that  $z = 2 + 3i$  is a root of the equation  $z^3 - 2z^2 + pz + q = 0$ , where  $p$  and  $q$  are real.

(i) Find the other two roots of the equation. 2

Since  $z = 2 + 3i$  is a root then so is  $z = 2 - 3i$  (since coefficients are real)

Let roots be  $z(\alpha) = 2 + 3i$ ,  $z(\beta) = 2 - 3i$  and  $z = \gamma$

$\therefore \alpha + \beta + \gamma = 2$  (sum of roots)

$$(2 + 3i) + (2 - 3i) + \gamma = 2$$

$$\gamma = -2$$

(ii) Find the values of  $p$  and  $q$ . 2

$$\alpha\beta\gamma = -q \quad \dots\dots\dots (1)$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = p \quad \dots\dots\dots (2)$$

$$\text{from (1)} \quad (2 + 3i)(2 - 3i)(-2) = -q$$

$$\Rightarrow q = 26$$

$$\text{from (2)} \quad (2 + 3i)(2 - 3i) + (2 + 3i)(-2) + (2 - 3i)(-2) = p$$

$$\Rightarrow p = 5$$

$$\therefore p = 5 \text{ and } q = 26$$

- (c) (i) If  $z = \cos \theta + i \sin \theta$ , use De Moivre's Theorem to prove that  $z^n + z^{-n} = 2 \cos n\theta$ . 2

$$\begin{aligned} z^n + z^{-n} &= (\cos \theta + i \sin \theta)^n + (\cos \theta + i \sin \theta)^{-n} \\ &= (\cos n\theta + i \sin n\theta) + (\cos(-n\theta) + i \sin(-n\theta)) \quad (\text{by De Moivre's Theorem}) \\ &= (\cos n\theta + i \sin n\theta) + (\cos n\theta - i \sin n\theta) \\ &= 2 \cos n\theta \end{aligned}$$

- (ii) Hence deduce that  $\cos \theta \cos 2\theta = \frac{1}{2}(\cos 3\theta + \cos \theta)$ . 3

$$\begin{aligned} \text{Now } \cos n\theta &= \frac{z^n + z^{-n}}{2} \\ \cos \theta \cos 2\theta &= \left( \frac{z + z^{-1}}{2} \right) \left( \frac{z^2 + z^{-2}}{2} \right) \\ &= \frac{z^3 + z^{-1} + z + z^{-3}}{4} \\ &= \frac{1}{4} \left( (z^3 + z^{-3}) + (z + z^{-1}) \right) \\ &= \frac{1}{4} (2 \cos 3\theta + 2 \cos \theta) \\ \therefore \cos \theta \cos 2\theta &= \frac{1}{2} (\cos 3\theta + \cos \theta) \end{aligned}$$

**Question 2: (15 Marks) (START A NEW PAGE)****Marks**

- (a) (i) Solve  $\alpha^2 = 5 - 12i$ , expressing both answers in the form  $a + ib$  where  $a$  and  $b$  are real numbers.

**3**

Let  $\alpha = a + ib$  where  $a$  and  $b$  are real

$$(a + ib)^2 = 5 - 12i$$

$$a^2 - b^2 + 2iab = 5 - 12i$$

$$a^2 - b^2 = 5 \quad \dots\dots\dots(1)$$

$$ab = -6 \quad \dots\dots\dots(2)$$

By inspection  $a = 3$  and  $b = -2$  or  $a = -2$  and  $b = 3$

$$\therefore \alpha = 3 - 2i \text{ or } -3 + 2i$$

- (ii) Hence solve  $z^2 - (1 - 4i)z - (5 - i) = 0$ , expressing both answers in the form  $x + iy$  where  $x$  and  $y$  are real numbers.

**3**

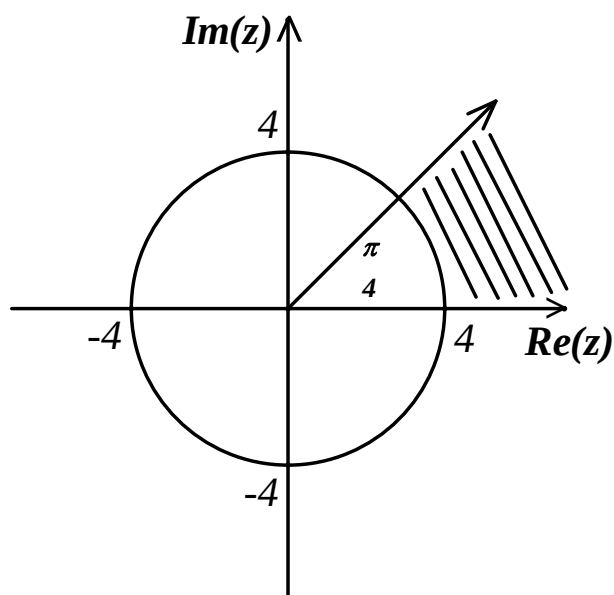
$$z = \frac{(1 - 4i) \pm \sqrt{(1 - 4i)^2 - 4(1)(- (5 - i))}}{2}$$

$$= \frac{(1 - 4i) \pm \sqrt{5 - 12i}}{2}$$

$$= \frac{(1 - 4i) \pm (3 - 2i)}{2}$$

$$z = 3 - 2i \text{ or } -1 - i$$

- (b) Sketch, on an Argand Diagram, the intersection of the regions  $|z| \geq 4$  and  $0 \leq \arg(z) \leq \frac{\pi}{4}$ .

**3**

(c) If  $z = \frac{1+i}{1-i}$  and  $w = \frac{\sqrt{2}}{1-i}$  are two complex numbers.

(i) Find the modulus and arguments of  $z$  and  $w$ .

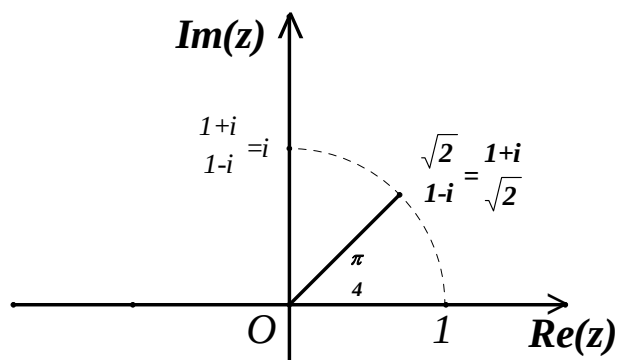
2

$$\begin{aligned}
 |z| &= \frac{|1+i|}{|1-i|} & \arg z &= \arg\left(\frac{1+i}{1-i}\right) \\
 &= \frac{\sqrt{2}}{\sqrt{2}} & &= \arg(1+i) - \arg(1-i) \\
 &= 1 & &= \frac{\pi}{4} - \left(-\frac{\pi}{4}\right) \\
 & & &= \frac{\pi}{2}
 \end{aligned}$$

$$\begin{aligned}
 |w| &= \frac{|\sqrt{2}|}{|1-i|} & \arg w &= \arg\left(\frac{\sqrt{2}}{1-i}\right) \\
 &= \frac{\sqrt{2}}{\sqrt{2}} & &= \arg(\sqrt{2}) - \arg(1-i) \\
 &= 1 & &= 0 - \left(-\frac{\pi}{4}\right) \\
 & & &= \frac{\pi}{4}
 \end{aligned}$$

(ii) Plot the points representing  $z$ ,  $w$  and  $z + w$  on an Argand Diagram.

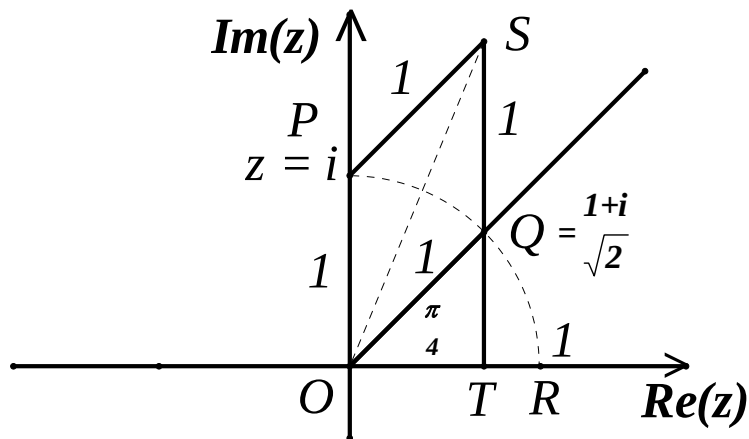
2



(iii) Hence show that  $\tan\left(\frac{3\pi}{8}\right) = 1 + \sqrt{2}$

2

$$\begin{aligned}
 z &= \frac{1+i}{1-i} = \operatorname{cis} \frac{\pi}{2} = i \\
 w &= \frac{\sqrt{2}}{1-i} = \operatorname{cis} \frac{\pi}{4} = \frac{1}{\sqrt{2}}(1+i)
 \end{aligned}$$



$$\angle POR = \frac{\pi}{2} \quad \text{and} \quad \angle QOR = \frac{\pi}{4} \Rightarrow \angle POQ = \frac{\pi}{4}$$

$$\angle SOQ = \frac{\pi}{8} \quad (\text{POQS is a rhombus and diagonals bisect opposite angles})$$

$$\angle SOR = \frac{3\pi}{8}$$

$$\tan(\angle SOR) = \frac{ST}{OT}$$

$$OT = QT \quad (\triangle OTQ \text{ is isosceles})$$

$$\frac{QT}{1} = \sin \frac{\pi}{4}$$

$$QT = OT = \frac{1}{\sqrt{2}}$$

$$ST = 1 + \frac{1}{\sqrt{2}}$$

$$\begin{aligned} \tan \frac{3\pi}{8} &= \frac{1 + \frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}}} \\ &= 1 + \sqrt{2} \end{aligned}$$

**Question 3: (15 Marks) (START A NEW PAGE)****Marks**

- (a) (i) Express  $1 + i\sqrt{3}$  in mod-arg form.

**2**

$$1 + i\sqrt{3} = 2\text{cis} \frac{\pi}{3}$$

- (ii) Hence evaluate  $(1 + i\sqrt{3})^8$ , expressing your answer in the form  $x + iy$  where  $x$  and  $y$  are real numbers and not trigonometric expressions.

**2**

$$\begin{aligned}(1 + i\sqrt{3})^8 &= \left(2\text{cis} \frac{\pi}{3}\right)^8 \\&= 256\text{cis} \frac{8\pi}{3} \\&= 256\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right) \\&= -128 + 128i\sqrt{3}\end{aligned}$$

- (b) Solve  $\bar{z} - \frac{12}{z} = 1 - i$ , expressing your answer in the form  $a + ib$  where  $a$  and  $b$  are real numbers.

**4**

Let  $z = a + ib$  where  $a$  and  $b$  are real

$$a - ib - \frac{12}{a + ib} = 1 - i$$

$$(a + ib)(a - ib) - 12 = (1 - i)(a + ib)$$

$$a^2 + b^2 - 12 = a + ib - ia + b$$

$$a^2 + b^2 - 12 = a + b + i(b - a)$$

on equating real and imaginary parts

$$a^2 + b^2 - 12 = a + b \quad \dots\dots\dots(1)$$

$$b - a = 0 \quad \dots\dots\dots(2)$$

substitute (2) into (1)

$$2a^2 - 12 = 2a$$

$$a^2 - a - 6 = 0$$

$$(a - 3)(a + 2) = 0$$

$$a = -2 \text{ or } 3$$

$$a = -2 \Rightarrow b = 2$$

$$a = 3 \Rightarrow b = -3$$

$$z = -2 - 2i \quad \text{or} \quad 3 + 3i$$

- (c) The point  $Z$  represents the complex number  $z$  in the complex plane. If  $\operatorname{Re}\left(\frac{z-ib}{z-a}\right) = 0$ , where  $a$  and  $b$  are real, prove that the locus of  $Z$  is a circle and give its centre and radius.

$$\begin{aligned}\frac{z-ib}{z-a} &= \frac{x+iy-ib}{x+iy+a} \\ &= \frac{x+i(y-b)}{(x+a)+iy} \times \frac{(x+a)-iy}{(x+a)-iy} \\ &= \frac{[x(x-a)+y(y-b)]+i[-xy+(x-a)(y-b)]}{(x+a)^2+y^2}\end{aligned}$$

$$\text{If } \operatorname{Re}\left(\frac{z-ib}{z-a}\right) = 0 \text{ then } x(x-a)+y(y-b)=0$$

$$x^2-ax+y^2-yb=0$$

$$\left(x-\frac{a}{2}\right)^2 + \left(y-\frac{b}{2}\right)^2 = \frac{a^2+b^2}{4}$$

Therefore locus is a circle with centre  $\left(\frac{a}{2}, \frac{b}{2}\right)$  and radius  $\frac{\sqrt{a^2+b^2}}{2}$

- (d) If  $z = r(\cos \theta + i \sin \theta)$ , prove that  $\frac{z}{z^2+r^2}$  is real and express its value in terms of  $\theta$ .

$$\begin{aligned}\frac{z}{z^2+r^2} &= \frac{r(\cos \theta + i \sin \theta)}{\{r(\cos \theta + i \sin \theta)\}^2 + r^2} \quad (z \neq ir) \\ &= \frac{r(\cos \theta + i \sin \theta)}{r^2(\cos 2\theta + i \sin 2\theta) + r^2} \\ &= \frac{(\cos \theta + i \sin \theta)}{r(\cos 2\theta + i \sin 2\theta + 1)} \\ &= \frac{1}{r} \left( \frac{\cos \theta + i \sin \theta}{2 \cos^2 \theta + 2i \sin \theta \cos \theta} \right) \\ &= \frac{1}{2r \cos \theta} \left( \frac{\cos \theta + i \sin \theta}{\cos \theta + i \sin \theta} \right)\end{aligned}$$

$$\frac{z}{z^2+r^2} = \frac{1}{2r \cos \theta}$$

Therefore  $\frac{z}{z^2+r^2}$  is real and its value is  $\frac{1}{2r \cos \theta}$

**Question 4: (15 Marks) (START A NEW PAGE)**

**Marks**

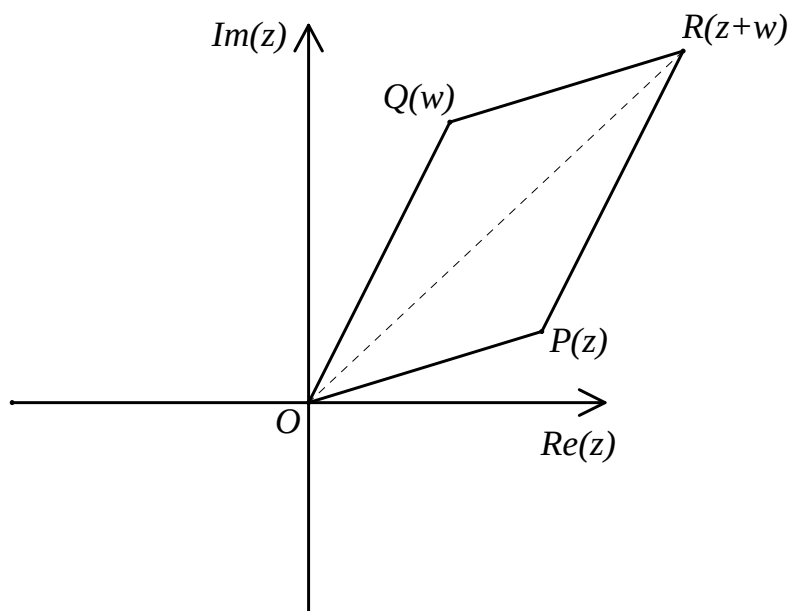
- (a) If  $\alpha$  and  $\beta$  are two complex numbers and  $|\alpha|=1$ , show that  $\left| \frac{\alpha\bar{\beta}-1}{\alpha-\beta} \right| = 1$ .

**3**

$$\begin{aligned}
 \left| \frac{\alpha\bar{\beta}-1}{\alpha-\beta} \right| &= \frac{|\alpha\bar{\beta}-1|}{|\alpha-\beta|} \\
 &= \frac{|\alpha\bar{\beta}-1|}{|\bar{\alpha}-\bar{\beta}|} \quad (\text{since } |z| = |\bar{z}|) \\
 &= \frac{1}{|\alpha|} \frac{|\alpha\bar{\beta}-1|}{|\bar{\alpha}-\bar{\beta}|} \quad (\text{since } |\alpha|=1) \\
 &= \frac{|\alpha\bar{\beta}-1|}{|\alpha\bar{\alpha}-\alpha\bar{\beta}|} \\
 &= \frac{|\alpha\bar{\beta}-1|}{|1-\alpha\bar{\beta}|} \quad (\text{since } |\alpha\bar{\alpha}|=1) \\
 &= 1
 \end{aligned}$$

- (b) (i) The points  $P$ ,  $Q$  and  $R$  represent the complex numbers  $z$ ,  $w$  and  $z+w$  respectively on an Argand diagram, where  $\arg(w) > \arg(z)$  and  $0 < \arg(z) < \frac{\pi}{2}$ . Plot the positions of points  $P$ ,  $Q$  and  $R$  on an Argand diagram.

**1**



- (ii) If  $O$  is the origin, what type of quadrilateral is  $OPRQ$  given that  $\left| \frac{z-w}{z+w} \right| = 1$ ? 1

$$\left| \frac{z-w}{z+w} \right| = 1$$

$$|z-w| = |z+w|$$

$$\therefore OR = PQ$$

Since  $OPRQ$  is a parallelogram by construction of  $z+w$  then  $OPRQ$  is a rectangle (parallelogram with equal diagonals) or possibly a square if  $|z| = |w|$ .

- (iii) Hence, what can be stated about the expression  $\frac{w}{z}$ ? 2

Since  $OPRQ$  is a rectangle  $\angle QOP = \frac{\pi}{2}$  (angles at vertex in rectangle =  $\frac{\pi}{2}$ )

$$\text{Now } \arg\left(\frac{w}{z}\right) = \arg(w) - \arg(z)$$

$$\text{But } \arg(w) - \arg(z) = \angle QOP$$

$$\therefore \arg\left(\frac{w}{z}\right) = \frac{\pi}{2}$$

$$\therefore \frac{w}{z} \text{ is a positive pure imaginary number}$$

- (c) (i) Write  $-1$  in mod-arg form and hence express both complex roots of  $z^3 = -1$  in mod-arg form. 3

$$-1 = \text{cis } \pi$$

$$\text{Let } z = r \text{cis } \theta$$

$$z^3 = r^3 \text{cis } 3\theta$$

$$r^3 \text{cis } 3\theta = \text{cis}(\pi + 2k\pi) \quad k = 0, 1, 2$$

$$r = 1, 3\theta = \pi + 2k\pi$$

$$\theta = \frac{(2k+1)\pi}{3}$$

$$\theta = \frac{\pi}{3}, \pi, \frac{5\pi}{3}$$

$$\text{Complex roots are } z = \text{cis } \frac{\pi}{3} \text{ and } \text{cis } \frac{5\pi}{3}$$

- (ii) If  $z = \lambda$  is either of the two complex roots of  $z^3 = -1$ , prove that  $\lambda^2 - \lambda + 1 = 0$ . 2

$$z^3 = -1$$

$$z^3 + 1 = 0$$

$$(z+1)(z^2 - z + 1) = 0$$

$$\text{when } z = \lambda \quad (\lambda \in \mathbb{C})$$

$$(\lambda+1)(\lambda^2 - \lambda + 1) = 0$$

but  $\lambda \neq -1$  since  $\lambda$  is complex

$$\therefore \lambda^2 - \lambda + 1 = 0$$

- (iii) Hence evaluate  $(\lambda^2 + \lambda + 1)^8 + (\lambda^2 + \lambda - 1)^8$ . 3

$$\begin{aligned} (\lambda^2 + \lambda + 1)^8 + (\lambda^2 + \lambda - 1)^8 &= (2\lambda)^8 + (2\lambda^2)^8 \text{ since } \lambda^2 + 1 = \lambda \text{ and } \lambda - 1 = \lambda^2 \\ &= 256(\lambda^8 + \lambda^{16}) \\ &= 256\{(\lambda^3)^2 \lambda^2 + (\lambda^3)^5 \lambda\} \\ &= 256(\lambda^2 - \lambda) \text{ since } \lambda^3 = -1 \\ &= 256(-1) \\ &= -256 \end{aligned}$$



THIS IS THE END OF THE EXAMINATION PAPER



Alternative

**Question 4: (15 Marks) (START A NEW PAGE)**

**Marks**

- (a) Find the greatest value of  $|z|$  if the complex number  $z$  satisfies the condition  $\left|z - \frac{4}{z}\right| = 8$ . **4**
- (b) (i) Find the two complex cube roots of unity expressing them in mod-arg form. **3**
- (ii) If  $\omega$  is one of the complex cube roots of unity prove that  $1 + \omega + \omega^2 = 0$ . **2**
- (iii) Evaluate  $(1 + \omega - \omega^2)^8 + (1 - \omega + \omega^2)^8$ . **3**
- (iv) Prove that if  $n$  is an integer, then  $1 + \omega^n + \omega^{2n} = 3$  or  $0$ , depending on whether  $n$  is or is not a multiple of 3. **3**

**Extra Questions**

- (a) On the Argand Diagram provided and with the aid of ruler, protractor and a pair of compasses, show the position of the complex numbers represented by
- (i)  $\bar{z}_1$  **1**
- (ii)  $-z_2$  **1**
- (iii)  $z_1 - z_2$  **1**
- (iv)  $iz_1$  **1**
- (v)  $iz_1 + z_2$  **1**

Clearly show all relevant lengths and angles.

If  $\omega$  is a complex root of  $\omega^8 = 1$ , prove that  $\omega + \omega^7$  is real.